Material Safety Data Sheet
#202 Thermal Guard CODE

Issued: 8/21/95 Revised: 6/3/08

SECTION 1: CHEMICAL PRODUCT AND COMPANY IDENTIFICATION

Frontier Performance Lubricants, Inc.
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800-807-4496

24 Hour Emergency Telephone Number
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PRODUCT NAME:	**THERMAL GUARD PARA-SYN GEAR OIL 85W/140**
CHEMICAL NAME:	**PETROLEUM LUBRICATING OIL**
CAS #:	**MIXTURE**
PRODUCT CATEGORY:	**GEAR OIL**
MSDS CODE:	**202**
ISSUED:	**8/21/95**
REVISED:	**6/3/08**

SECTION 2: COMPOSITION / INFORMATION ON INGREDIENTS

PRODUCT	**CAS NO. OF COMPONENTS**	
LUBRICATING BASE STOCK(S)	CAS# 64742-57-0	>55%
PROPRIETARY ADDITIVES	MIXTURE	<15%

PLEASE NOTE THAT THE CHEMICAL IDENTITY OF SOME OR ALL OF THE ABOVE INGREDIENTS IS CONFIDENTIAL INFORMATION AND MAY BE WITHHELD AS PERMITTED BY 29CFR 1910.1200 AND VARIOUS STATE RIGHT TO KNOW LAWS.

HAZARDOUS MATERIALS IDENTIFICATION SYSTEM (HMIS):

HEALTH	FLAMMABILITY	REACTIVITY
1	1	0

EXPOSURE LIMITS:

OIL MIST: ACGIH: TLV/TWA - 5 MG/M3, TLV/STEL - 10MG/M3
OSHA: PEL/TWA - 5 MG/M3, PEL/CEILING - NONE

SECTION 3: HAZARD IDENTIFICATION

VARIABLE AMONG INDIVIDUALS:
HEALTH STUDIES HAVE SHOWN THAT MANY PETROLEUM HYDROCARBONS AND SYNTHETIC LUBRICANTS POSE POTENTIAL HUMAN HEALTH RISKS WHICH MAY VARY FROM PERSON TO PERSON. AS A PRECAUTION, EXPOSURE TO LIQUIDS, VAPORS, MISTS OR FUMES SHOULD BE MINIMIZED.

EFFECTS OF OVEREXPOSURE (SIGNS AND SYMPTOMS OF EXPOSURE):
SKIN:
THIS MATERIAL MAY BE SLIGHTLY IRRITATING TO THE SKIN. IT MAY CAUSE SKIN IRRITATION AND/OR ALLERGIC SKIN REACTION. PROLONGED AND REPEATED CONTACT MAY LEAD TO VARIOUS SKIN DISORDERS SUCH AS DERMATITIS, OIL ACNE, OR FOLLICULITIS. PERSONS WITH PRE-EXISTING SKIN DISORDERS MAY BE MORE SUSCEPTIBLE TO THE EFFECTS OF THIS MATERIAL. BASED ON DATA FROM COMPONENTS OR SIMILAR MATERIALS.

EYE:

THIS PRODUCT MAY CAUSE MILD EYE IRRITATION. DIRECT CONTACT WITH THE LIQUID OR EXPOSURE TO VAPORS OR MISTS MAY CAUSE STINGING, TEARING AND REDNESS. BASED ON DATA FROM COMPONENTS OR SIMILAR MATERIALS.

INGESTION:
INGESTION OF EXCESSIVE AMOUNTS MAY CAUSE IRRITATION OF THE DIGESTIVE TRACT, NAUSEA AND VOMITING. ASPIRATION INTO THE LUNGS DURING INGESTION OR VOMITING MAY CAUSE MILD TO SEVERE PULMONARY INJURY AND POSSIBLE DEATH. BASED ON DATA FROM COMPONENTS OR SIMILAR MATERIALS.

RESPIRATORY:
INHALATION OF VAPORS OR MISTS MAY BE HARMFUL. BREATHING HIGH CONCENTRATIONS OF VAPORS OR MISTS MAY CAUSE IRRITATION OF THE NOSE AND THROAT. BASED ON DATA FROM COMPONENTS OR SIMILAR MATERIALS.

SECTION 4: FIRST AID MEASURES

EYE CONTACT:
IMMEDIATELY FLUSH EYES WITH LARGE AMOUNTS OF WATER FOR AT LEAST 15 MINUTES. GET MEDICAL ATTENTION IF IRRITATION PERSISTS OR DEVELOPS.

SKIN:
IN CASE OF SKIN CONTACT, REMOVE ANY CONTAMINATED CLOTHING AND WIPE EXCESS OFF. WASH SKIN WITH SOAP AND WATER OR A WATERLESS HAND CLEANER FOLLOWED BY SOAP AND WATER. LAUNDER OR DRY-CLEAN CLOTHING BEFORE REUSE AND DISCARD SHOES AND OTHER LEATHER ARTICLES SATURATED WITH THE MATERIAL. GET MEDICAL ATTENTION IF IRRITATION PERSISTS OR DEVELOPS.

INHALATION:
REMOVE EXPOSED PERSON TO FRESH AIR IF ADVERSE EFFECTS ARE OBSERVED. IF BREATHING IS LABORED, ADMINISTER OXYGEN. IF BREATHING HAS STOPPED, APPLY ARTIFICIAL RESPIRATION. SEEK IMMEDIATE MEDICAL ATTENTION.

INGESTION:
DO NOT INDUCE VOMITING, BECAUSE MATERIAL CAN ENTER THE LUNGS AND CAUSE SEVERE DAMAGE. DRINK 4 TO 8 OUNCES OF WATER AND SEEK IMMEDIATE MEDICAL ATTENTION. IF VOMITING OCCURS, OR IF COMATOSE OR LETHARGIC, PLACE ON THE LEFT SIDE WITH THE HEAD DOWN. SEEK IMMEDIATE MEDICAL ATTENTION.

SECTION 5: FIRE FIGHTING MEASURES

FLASH POINT:	330 DEG. F COC (MIN).
UPPER FLAMMABILITY LIMIT:	NOT DETERMINED
LOWER FLAMMABILITY LIMIT:	NOT DETERMINED

EXTINGUISHING MEDIA AND FIRE FIGHTING PROCEDURES:
CARBON DIOXIDE, DRY CHEMICAL, FOAM OR WATER FOG. WATER SPRAY CAN BE USED TO COOL AND PROTECT CONTAINERS EXPOSED TO HEAT AND FLAME. PRODUCT WILL FLOAT ON SURFACE OF WATER.

SPECIAL FIREFIGHTING PROCEDURES: RECOMMEND WEARING POSITIVE PRESSURE SELF-CONTAINED BREATHING APPARATUS. DO NOT ENTER CONFINED FIRE-SPACE WITHOUT FULL BUNKER GEAR (HELMET WITH FACE SHIELD, BUNKER COATS, GLOVES AND RUBBER BOOTS). TOXIC FUMES, GASES OR VAPORS MAY EVOLVE ON BURNING. VAPORS MAY BE HEAVIER THAN AIR AND MAY TRAVEL ALONG THE GROUND TO A DISTANT IGNITION SOURCE AND FLASH BACK.

UNUSUAL FIRE & EXPLOSIVE HAZARDS: SLIGHTLY COMBUSTIBLE WHEN HEATED ABOVE FLASH POINT. WILL RELEASE FLAMMABLE VAPORS WHICH CAN BURN IN OPEN OR BE EXPLOSIVE IN CONFINED SPACES IF EXPOSED TO IGNITION.

DECOMPOSITION PRODUCTS UNDER FIRE CONDITION:
FUMES, SMOKE, CARBON MONOXIDE, ALDEHYDES AND OTHER PRODUCTS OF INCOMPLETE COMBUSTION. OXIDES NITROGEN, PHOSPHORUS AND SULFUR WILL BE FORMED.

SECTION 6: ACCIDENTAL RELEASE MEASURES

STEPS TO BE TAKEN IN CASE MATERIAL IS RELEASED OR SPILLED:

REMOVE SOURCES OF IGNITION. PERSONNEL PROTECTIVE EQUIPMENT MUST BE WORN(SEE SECTION VIII). VENTILATE SPILL AREA IF IN CONFINED OR POORLY VENTILATED AREA. PICK UP FREE LIQUID WITH EXPLOSION PROOF OR HAND PUMP FOR RECYCLE AND/OR DISPOSAL IF IT CAN BE DONE SAFELY. RESIDUAL MATERIAL CAN BE ABSORBED ON INERT MATERIAL. MINIMIZE SKIN CONTACT. KEEP PRODUCT OUT OF SEWERS AND WATERCOURSES BY DIKING OR IMPOUNDING. ADVISE AUTHORITIES IF PRODUCT HAS ENTERED OR MAY ENTER SEWERS, WATERCOURSES, OR EXTENSIVE LAND AREAS. ASSURE CONFORMITY WITH APPLICABLE GOVERNMENTAL REGULATIONS

WASTE DISPOSAL:
WASTE DISPOSAL SHOULD BE IN COMPLIANCE WITH ALL FEDERAL, STATE, AND LOCAL LAWS.

SECTION 7: HANDLING AND STORAGE

STORE IN A COOL DRY LOCATION. KEEP AWAY FROM POTENTIAL SOURCES IF IGNITION. KEEP AWAY FROM INCOMPATIBLE MATERIALS. OPEN CONTAINER IN A WELL VENTILATED AREA. AVOID GENERATING MISTS WHEN HANDLING. AVOID BREATHING VAPORS, OR PROLONGED OR REPEATED SKIN CONTACT. WASH THROUGHLY AFTER HANDLING. REMOVE ANY CONTAMINATED CLOTHING AND LAUNDER BEFORE REUSE.

"EMPTY" CONTAINER WARNING:
"EMPTY" CONTAINERS RETAIN RESIDUE (LIQUID AND/OR VAPOR) AND CAN BE DANGEROUS. **DO NOT PRESSURIZE, CUT, WELD, BRAZE SOLDER, DRILL, GRIND, OR EXPOSE SUCH CONTAINERS TO HEAT, FLAME, SPARKS, STATIC ELECTRICITY, OR OTHER SOURCES OF IGNITION; THEY MAY EXPLODE AND CAUSE INJURY OR DEATH.** DO NOT ATTEMPT TO CLEAN SINCE RESIDUE IS DIFFICULT TO REMOVE. "EMPTY" DRUMS SHOULD BE COMPLETELY DRAINED, PROPERLY BUNGED AND PROMPTLY RETURNED TO A DRUM RECONDITIONER. ALL OTHER CONTAINERS SHOULD BE DISPOSED OF IN AN ENVIRONMENTALLY SAFE MANNER AND IN ACCORDANCE WITH GOVERNMENTAL REGULATIONS.

SECTION 8: EXPOSURE CONTROLS - PERSONAL PROTECTION

VENTILATION:
PROVIDE VENTILATION SUFFICIENT TO PREVENT EXCEEDING RECOMMENDED EXPOSURE LIMIT. USE LOCAL EXHAUST TO CAPTURE VAPOR, MISTS OR FUMES, IF NECESSARY. NO SMOKING, FLAME OR OTHER IGNITION SOURCES.

RESPIRATORY PROTECTION:
UNDER NORMAL USE CONDITIONS, RESPIRATOR IS NOT USUALLY REQUIRED. USE NIOSH/MSHA APPROVED DISPOSABLE DUST/MIST MASK OR AIR-PURIFYING RESPIRATOR FOR ORGANIC VAPORS AND PARTICULATES IF THE RECOMMENDED EXPOSURE LIMIT IS EXCEEDED. USE SELF-CONTAINED BREATHING APPARATUS FOR ENTRY INTO CONFINED SPACE, FOR OTHER POORLY VENTILATED AREAS AND FOR LARGE SPILL CLEAN-UP SITES.

PROTECTIVE GLOVES:
USE CHEMICAL RESISTANT GLOVES TO AVOID PROLONGED OR REPEATED SKIN CONTACT. NITRILE GLOVES ARE RECOMMENDED.

EYE PROTECTION:
USE SPLASH GOGGLES OR FACE SHIELD WHEN EYE CONTACT MAY OCCUR.

OTHER PROTECTIVE EQUIPMENT:
USE CHEMICAL RESISTANT BOOTS AND APRONS OR OTHER IMPERVIOUS CLOTHING, IF NEEDED, TO AVOID CONTAMINATING REGULAR

CLOTHING. LONG SLEEVE SHIRT IS RECOMMENDED.

WORK PRACTICES:

KEEP CONTAINERS CLOSED WHEN NOT IN USE. DO NOT STORE NEAR HEAT, SPARKS, FLAME OR STRONG OXIDANTS.

PERSONAL HYGIENE:
MINIMIZE BREATHING VAPOR, MIST OR FUMES. AVOID PROLONGED OR REPEATED CONTACT WITH SKIN. REMOVE CONTAMINATED CLOTHING; LAUNDER BEFORE REUSE. REMOVE CONTAMINATED SHOES AND THOROUGHLY CLEAN BEFORE RE-USE; DISCARD IF OIL-SOAKED. CLEANSE SKIN THOROUGHLY AFTER CONTACT, BEFORE BREAKS AND MEALS, AND AT END OF WORK PERIOD.

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

IX. PHYSICAL DATA

SPECIFIC GRAVITY: 0.88 - 0.90 @ 15.6C	**VAPOR PRESSURE:** NEGLIGIBLE AT AMBIENT
% VOLATILE BY VOLUME: NEGLIGIBLE	**VAPOR DENSITY (AIR=1)** HEAVIER THAN AIR
SOLUBILITY IN WATER: NEGLIGIBLE	**EVAPORATION RATE (n-BUTYL ACETATE =1)** NEGLIGIBLE

SECTION 10: STABILITY AND REACTIVITY

STABILITY: STABLE UNDER NORMAL CONDITIONS OF STORAGE AND HANDLING.
HAZARDOUS POLYMERIZATION: WILL NOT OCCUR
AVOID CONTACT WITH: STRONG OXIDIZING AGENTS

SECTION 11: TOXICOLOGICAL INFORMATION

REPORTABLE QUANTITY (RQ), EPA REGULATION 40 CFR 302 (CERCLA SECTION 102):
NO RQ FOR PRODUCT KNOWN TO BE GREATER THAN 1% OR 01% (CARCINOGEN).

SARA TITLE III:
SECTION 313 REPORTABLE INGREDIENTS:
THIS PRODUCT IS NOT KNOWN TO CONTAIN GREATER THAN 1% OR 0.1% (CARCINOGEN) OF ANY CHEMICAL SUBSTANCES LISTED UNDER SARA SECTION 313.

SECTION 311/312:
IMMEDIATE HEALTH

SECTION 12: TRANSPORTATION INFORMATION

DOT PROPER SHIPPING NAME FOR NON-BULK: NOT APPLICABLE
DOT HAZARD CLASS: NOT REGULATED
DOT UN NUMBER: NOT APPLICABLE

SECTION 13: OTHER INFORMATION

THE INFORMATION AND RECOMMENDATIONS HEREIN ARE, TO THE BEST OF OUR KNOWLEDGE AND BELIEF, ACCURATE AND RELIABLE AS OF THE DATE ISSUED. FRONTIER DOES NOT WARRANT OR GUARANTEE THEIR ACCURACY OR RELIABILITY, AND SHALL NOT BE LIABLE FOR ANY LOSS OR DAMAGE ARISING OUT OF THE USE THEREOF.

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